# INFINITE TUTORIAL - OFFICIAL EXAMINATION PAPER

## CBSE - CLASS 10 SCIENCE: PHYSICS (ELECTRICITY & CIRCUITS)

Max Marks: 40 | Duration: 60 Minutes | Subject Code: PHY10

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Candidate Name: _____ Roll Number: _____ Date: _____

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General Instructions: Draw neat circuit diagrams. Step marks awarded for numerical formulas.

SECTION A - CONCEPTUAL (1 Mark Each)

Q1. State Ohmâ s Law and write its mathematical formula.

Q2. Name the physical quantity whose SI unit is Ohm-metre (Resistivity).

SECTION B - NUMERICAL PROBLEMS (2 Marks Each)

Q3. A potential difference of 220V is applied across a resistance of 100 ohms. Calculate current.

Q4. Why is tungsten used almost exclusively for filaments of electric lamps?

SECTION C - APPLICATION & CIRCUITS (3 Marks Each)

Q5. Three resistors 2 ohm, 3 ohm, and 6 ohm are connected in parallel. Find equivalent resistance.

Q6. Compute the heat generated while transferring 96,000 coulombs of charge in 1 hour through 50V.

SECTION D - EXPERIMENTAL ANALYSIS (5 Marks Each)

Q7. Derive the formula for Jouleâ s Law of Heating H = I^2 R t.

Q8. State reasons why series arrangement is not used for domestic circuits.

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*** End of Question Paper - Certified by Infinite Tutorial Academic Directorate ***